

Exam Fall 2025

1. Consider two independent random variables $X \sim N(10, 0.25)$ and $Y \sim N(5, 0.01)$

Let $Z = X - 4 * Y$

What is the distribution of Z ?

*Here the **first** parameter represents the mean and the **second** parameter represents variance of the normal distribution.*

- A) $N(5, 1)$
- B) $N(5, 5)$
- C) $N(5, 0.9)$
- D) $N(-10, 0.09)$

Correct answer: D

$$E[Z] = E[X] - 4 * E[Y] = 10 - 4 * 5 = -10$$

$$\text{Var}[Z] = \text{Var}[X] - 4^2 * \text{Var}[Y] = 0.25 - 16 * 0.01 = 0.09$$

2. An experiment consists of measuring the time (in seconds) it takes for a computer to boot up. Assume the computer cannot boot instantly (it always takes some positive time). Which of the following best describes the sample space?

- A. $\{1, 2, 3, \dots, 60\}$
- B. All integer values between 0 and 60
- C. All real numbers greater than 0
- D. All real numbers between 0 and 60

Correct Answer: C

Time is a continuous variable, so the sample space consists of all positive real numbers (greater than 0). If the experiment imposes a maximum limit (e.g., 60 seconds), then the sample space would be all real numbers in the interval $(0, 60]$.

3. Let A and B be two independent events with $P(A) = 0.4$ and $P(B) = 0.5$. What is $P(A \cup B)$?

Here $\cup$ represents the union operator.

- A) 0.7
- B) 0.2

C) 0.9

D) 0.1

Correct answer: A

$$\begin{aligned} P(A \cup B) &= P(A) + P(B) - P(A \cap B) = P(A) + P(B) - P(A) * P(B) \text{ [due to independence]} \\ &= 0.4 + 0.5 - (0.4 * 0.5) = 0.7 \end{aligned}$$

4. If $P(X | Y) = 0.8$ and $P(Y) = 0.5$, what is $P(X \cap Y)$?

A) 0.4

B) 1.3

C) 0.3

D) 0.625

Correct answer: A

$$P(X | Y) = P(X \cap Y) / P(Y) \Rightarrow P(X \cap Y) = P(X | Y) * P(Y) = 0.8 * 0.5 = 0.4$$

5. For a discrete random variable X, the Probability Density Function (PDF) $P(X = x)$ must satisfy which condition?

A) Sum of $P(X = x) = 0$

B) Integral of $P(X = x) = 1$

C) Sum of $P(X = x) = 1$

D) $P(X = x) \leq 0$

Correct answer: C

6. A factory produces bulbs with a 10% defect rate. In a sample of 5 bulbs, what is the probability of exactly 1 defect?

A) 0.918

B) 0.328

C) 0

D) 0.2

Correct answer: B

$$\text{dbinom}(1, 5, 0.1) = 0.32805$$

7. For a random variable Z that follows the standard normal distribution, that is, $Z \sim N(0, 1)$, what is the value of $P(Z > 0)$?

- A) 0.0
- B) 0.5
- C) 1.0
- D) 0.398

Correct answer: B

$$\text{pnorm}(0) = 0.5$$

8. A sample of size $n = 10$ has a mean of $\bar{x} = 50$ and a sample standard deviation $s = 8$. Construct a 95% confidence interval for the population mean. Which of the following is closest to the correct interval?

- A. (44.3, 55.7)
- B. (45.3, 54.7)
- C. (46.0, 54.0)
- D. (38.6, 61.4)

Correct answer: A

$$\text{CI} = \text{mean}(\bar{x}) \pm t_{(1-\alpha/2) \times s/\sqrt{n}}$$

$$= [50 - \text{qt}(1 - 0.05/2, 9) * 8/\sqrt{10}, 50 + \text{qt}(1 - 0.05/2, 9) * 8/\sqrt{10}]$$

$$= [44.27, 55.73]$$

9. If the lifetime of a device follows an exponential distribution with $\lambda = 0.2$, what is the probability that it lasts more than 10 units of time?

- A. 0.999
- B. 0.864
- C. 4.539993e-05
- D. 0.135

Correct Answer: D

$$1 - \text{pexp}(10, 0.2) = 0.135$$

10. You are analyzing a dataset using Tukey's fences method. The first quartile (Q1) is 20, and the third quartile (Q3) is 40.

Which of the following values would be considered an outlier?

- A) 10
- B) 79
- C) 45
- D) 30

Correct answer: B

$$\text{IQR} = Q3 - Q1 = 40 - 20 = 20$$

$$\text{Lower} = Q1 - 1.5 * \text{IQR} = -10$$

$$\text{Upper} = Q3 + 1.5 * \text{IQR} = 70$$

11. Consider the following heights of 10 people:

`h <- c(168, 161, 167, 179, 184, 166, 198, 187, 191, 179)`

Which of the following regarding the 99% confidence interval for the population mean is correct?

- A) [169.2647, 186.7353]
- B) [168, 179]
- C) [165.4508, 190.5492]
- D) [165, 175]

Correct answer: C

`t.test(h, conf.level = 0.99)`

99 percent confidence interval:

[165.4508, 190.5492]

12. A Type I error occurs when:

Here H_0 is the null hypothesis.

- A) We reject H_0 when H_0 is true
- B) We accept H_0 when H_0 is false
- C) We reject H_0 when H_0 is false
- D) We accept H_0 when H_0 is true

Correct answer: A

Type 1 error is the false positive error (incorrect rejection of H_0).

13. A sample of size $n = 16$ has a mean $\bar{x} = 52$ and a sample standard deviation of 4. If we were to test the hypothesis $H_0: \mu = 50$ at 5% significance level, what is the value of the test statistic?

- A. 1.00
- B. 2.00
- C. 0.50
- D. 48.8

Correct answer: B

$$(52 - 50) / (4 / \sqrt{16}) = 2$$

14. What does the R command `qnorm(0.975)` return?

- A) The probability density at 0.975
- B) The cumulative probability at 0.975
- C) The value z such that $P(Z < z) = 0.975$
- D) A random number from the standard normal distribution

Correct answer: C

`Qnorm(0.975)` returns the 0.975th quantile

15. A researcher wants to test the effectiveness of a new teaching method. In the first study, she randomly assigns 50 students to one of two groups: one group uses the new method, and the other uses the traditional method. She then compares their final exam scores. In a second

study, she uses the same group of students, teaching them with both methods at different times, and then compares their scores.

Which statement best describes the difference between the two experimental designs?

- A. The first study is a paired experiment, while the second is randomized
- B. The first study is a randomized experiment, while the second is a paired experiment
- C. Both studies are randomized experiments
- D. None of the above

Correct answer: B.

16. For the given following data:

	Group 1	Group 2
Success	62	48
Failure	238	252
Total	300	300

- p_1 be the proportion of success in Group 1
- p_2 be the proportion of success in Group 2
- Assume $H_0 : p_1 = p_2$ and $H_1 : p_1 \neq p_2$

Perform hypothesis testing of proportions for the data at $\alpha = 0.01$ significance level.

Which of the following is correct?

- A. The p value is 0.001. Thus we reject H_0
- B. The p value is 0.005. Thus we reject H_0
- C. The p value is 0.139. Thus we accept H_0
- D. The p value is 0.139. Thus we reject H_0

Correct answer: C

`prop.test(x = c(62, 48), n = c(300, 300), correct = FALSE)`

17. What is a "residual" in linear regression?

- A) The predicted value minus the mean
- B) The difference between the observed value and the predicted value

- C) The slope of the line
D) The intercept

Correct answer: B

18. The joint probability distribution of two discrete random variables X and Y is given in the table below:

	$Y = 1$	$Y = 2$	$Y = 3$
$X = 1$	0.10	0.15	0.05
$X = 2$	0.20	0.25	0.25

Which of the following is the correct **marginal distribution of X** ?

- A. $P(X = 1) = 0.30, P(X = 2) = 0.70$
B. $P(X = 1) = 0.25, P(X = 2) = 0.50$
C. $P(X = 1) = 0.30, P(X = 2) = 0.40$
D. $P(X = 1) = 0.35, P(X = 2) = 0.65$

Correct answer: A

$$P(X = 1) = 0.10 + 0.15 + 0.05 = 0.30$$

$$P(X = 2) = 0.20 + 0.25 + 0.25 = 0.7$$

19. Which of the following statements about the Mann–Whitney U test applied on two samples of data is TRUE?

- A. It compares the variances of two independent samples.
B. It is a non-parametric alternative to the Student's t-test.
C. It requires both samples to be normally distributed.
D. It can only be used for paired data.

Correct answer: B

20. What does `summary(lm(y ~ x1 + x2, data = df))` output by default?

- A. Residual plots
B. Coefficients, standard errors, t-values, p-values, R^2 , adjusted R^2 , F-statistic...
C. Confidence intervals
D. None of the above

Answer: B

Confidence intervals require `confint()`; residual plots require `plot()` on the fitted model object.

21. Holt–Winters’ method is primarily used for time series data that exhibit:

- A. No trend and no seasonality
- B. Trend but no seasonality
- C. Seasonality but no trend
- D. Both trend and seasonality

Answer: D

Explanation: Holt–Winters (a.k.a. triple exponential smoothing) captures level, trend, and seasonality.

22. In a clinical trial of a cholesterol lowering medicine, 15 patient's cholesterol (in mmol/L) has been measured before treatment and 3 weeks after starting treatment. The data is given below.

```
chol_before <- c(9.1, 8.0, 7.7, 10.0, 9.6, 7.9, 9.0, 7.1, 8.3, 9.6, 8.2, 9.2, 7.3, 8.5, 9.5)
```

```
chol_after <- c(8.2, 6.4, 6.6, 8.5, 8.0, 5.8, 7.8, 7.2, 6.7, 9.8, 7.1, 7.7, 6.0, 6.6, 8.4)
```

Using statistical testing, can we say that the medicine has a positive effect in lowering the cholesterol levels at level of significance $\alpha = 0.001$?

- A. Yes, since the p value = $3.672e-06$ is less than α and H_0 is rejected
- B. No, since the p value = 0.002 is greater than α and H_0 is accepted
- C. Not enough information
- D. None of the above

Correct answer: A

```
t.test(chol_after, chol_before, paired = TRUE)
```

23. Given a sample of data (-0.34, 0.82, 0.50, x) and its mean = 0.5, what is the value of x?

- A. Cannot be determined.
- B. -0.48
- C. 1.2

D. 1.02

Correct answer: D

$$0.5 * 4 - (-0.34 + 0.82 + 0.50) = 1.02$$

24. A multiplicative time series model is appropriate when:

- A) The seasonal variation is constant
- B) The seasonal variation increases/decreases as the trend increases/decreases
- C) There is no trend
- D) The data has no seasonal component

Correct answer: B

25. Given that:

- standard deviation of sample 1 = 0.755
- standard deviation of sample 2 = 0.132
- covariance between sample 1 and sample 2 = 0.319

What is the value of the correlation between sample 1 and sample 2?

- A. 0.681
- B. 3.200
- C. 1.968
- D. 0.794

Correct answer: B

$$0.319 / (0.755 * 0.132) = 3.200$$

26. If X follows a Bernoulli distribution, and given that the mean, $E[X] = 0.3$, what is the value of the success probability (p) of this distribution?

- A) $p = 0.7$
- B) $p = 1$
- C) $p = 0.3$
- D) Cannot be determined

Correct answer: C

27. Which function calculates the probability $P(X = k)$ for a Binomial distribution?

- A) `dbinom()`
- B) `pbinom()`
- C) `qbinom()`
- D) `rbinom()`

Correct answer: A

28. Given $F(x) = 0.95$ for a standard normal distribution. What is the value of x ?

Here F is the cumulative distribution function (CDF).

- A. 1.644
- B. 0.835
- C. 0.5
- D. 1.959

Correct answer: A

`qnorm(0.95) = 1.644`

29. In a study of pollution in a water stream, the concentration of pollution is measured at 5 different locations. The concentration is modelled as a linear function of the distance to the pollution source.

`conc =  $\beta_0 + \beta_1$  distance_to_source`

In the code below, these distances and the average pollution concentrations are given:

`distance_to_source <- c(2, 4, 6, 8, 10)`

`conc <- c(11.5, 10.2, 10.3, 9.68, 9.32)`

The parameter estimates for the three unknown parameters in the linear regression model are:

- A. $\beta_0 = -0.244$, $\beta_1 = 11.66$, Error standard deviation (σ) = 1
- B. $\beta_0 = -0.244$, $\beta_1 = 11.66$, Error standard deviation (σ) = 0.8676

C. $\beta_0=11.66$, $\beta_1=-0.244$, Error standard deviation (σ) = 0.348

D. $\beta_0=0.365$, $\beta_1 = 0.055$, Error standard deviation (σ) = 0.348

Correct answer: C

```
m <- lm(conc ~ distance_to_source)
```

```
summary(m)
```